ADITYA UNIVERSITY

DATABASE MANAGEMENT SYSTEMS -2501IT05

(Common to CSE, IT, AIML & CSE (DS))

IE-2 Important Questions

UNIT-3

1. Explain the ER Model with neat diagrams.
2. Explain different types of attributes and relationships in the ER Model.
3. Explain Specialization, Generalization, Aggregation and Inheritance with examples.
4. Draw an ER diagram for a Library/Hospital/Banking System.
5. Explain the implementation of Primary Key, Foreign Key, UNIQUE, NOT NULL, and CHECK constraints in SQL with suitable examples.
6. What are integrity constraints? Explain Entity Integrity, Referential Integrity, and Domain Integrity constraints with suitable SQL examples.
7. Explain Nested and Correlated Subqueries with examples.
8. Explain different types of SQL Joins with suitable examples.
9. Differentiate updatable and non-updatable views with examples.

UNIT-4

1. Explain the purpose of Normalization and discuss 1NF, 2NF with suitable examples.
2. Explain 4NF, and 5NF with suitable examples.
3. Explain Functional Dependency and different types of dependencies with examples.
4. Compare 3NF and BCNF with suitable examples.
5. Explain Lossless Join and Dependency Preserving Decomposition.
6. Explain Transaction Management and ACID properties with examples.
7. Explain Transaction States with a neat state transition diagram.
8. Discuss conflict serializability and view serializability with suitable examples.
9. Explain Recoverability and Recovery Algorithms in DBMS.

UNIT-5

1. Explain B+ Tree and discuss search, insertion, and deletion algorithms with neat diagrams
2. Explain Primary, Secondary, and Clustered Indexes with suitable examples.
3. Compare Tree-Based Indexing and Hash-Based Indexing
4. Explain Hash-Based Indexing
5. What is Query Optimization? Explain the various Query Optimization Algorithms in detail with suitable examples.
6. What is Database Backup and Recovery? Explain the various Backup and Recovery Techniques in detail with suitable examples.